

Generation Interconnection System Impact Study Report

For Queue Project Q18-0951

Swansea 138 kV

March 01, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0951. The Interconnection Customer proposes a 230.0 MW solar photovoltaic facility interconnecting at the Swansea 138 kV substation in the Delaware area. The request was received on February 26, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 269 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,269,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sheffield Wind / Swansea 138kV line	Tombstone / Bixby Hill Substation 500kV	164.8	204	68.4
Sheffield Wind / Dolphus M Grainger 138kV line	Tombstone / Bixby Hill 500kV	137.4	222	68.4
Dolphus M Grainger / Aztec 138kV L	Bixby Hill / Tombstone 500kV Line	129.4	229	68.4
Clay / Swansea 138kV line	Sheffield Wind / Swansea 138kV	120.5	161	100.0
Blooming Prairie / Mill Cteek 138kV line	Star Peak / Hope Creek 345kV	107.5	214	5.1
Blooming Prairie / Glasgow 138kV	Hope Crk / Star Peak Substation 345kV line	105.9	209	5.1
Glasgow / Dell Creek Substation 138kV Line	Star Peak / Hope Creek 345kV	105.5	180	5.1
Clay / S. Macon Substation 138kV	Sheffield Wind / Swansa 138kV	100.8	184	100.0
Deer Vly / Chimayo 138kV	Aztec 345/138 kV Autotransformer	100.2	149	56.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sheffield Wind / Swansea 138kV line	Rebuild line with higher-capacity conductor	\$1,192,000	100% (\$1,192,000)
Dolphus M Grainger / Sheffield Wind 138kV	Rebuild line with higher-capacity conductor	\$4,249,000	100% (\$4,249,000)
Dolphus M Grainger / Aztec 138kV L	Rebuild line with higher-capacity conductor	\$2,682,000	100% (\$2,682,000)
Clay / Swansea 138kV line	Rebuild line with higher-capacity conductor	\$3,522,000	100% (\$3,522,000)
Blooming Prairie / Mill Cteek 138kV line	Reconductor line	\$2,414,000	64% (\$1,548,000)
Blooming Prairie / Glasgow 138kV	Reconductor line	\$6,078,000	64% (\$3,898,000)
Glasgow / Dell Creek Substation 138kV Line	Reconductor line	\$948,000	64% (\$608,000)
Clay / S. Macon Substation 138kV	Reconductor line	\$1,424,000	100% (\$1,424,000)
Deer Valley / Chimayo 138kV	Reconductor line	\$2,146,000	100% (\$2,146,000)
	Total allocated to project		\$21,269,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.